

Professional Summary

Full-Stack Developer and Software Engineering student at American International University-Bangladesh (AIUB) with hands-on experience designing, developing, and deploying production-grade web applications using Next.js, React, NestJS, TypeScript, PostgreSQL, Docker, and Socket.IO. Experienced in building scalable REST APIs, authentication systems, role-based access control (RBAC), real-time communication platforms, analytics dashboards, and business automation solutions. Published AI research author with strong interests in backend engineering, software architecture, distributed systems, and cloud-native application development.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, C, .NET

Frontend: Next.js, React.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, NestJS, REST APIs, JWT Authentication, Socket.IO

Databases: PostgreSQL, Prisma ORM, TypeORM

DevOps Infrastructure: Docker, Linux, Git, GitHub Actions, Vercel, Render, Neon

Machine Learning: Python, Pandas, NumPy, K-Nearest Neighbors (KNN), Data Analysis

Core Concepts: Software Engineering, System Design, Database Design, API Design, RBAC, Real-Time Systems, Software Architecture

Projects

ClassCircle — Full-Stack Educational Collaboration Platform

Jan 2025 – Present

Live Demo — GitHub Repository

- Built and deployed a production-grade SaaS platform using Next.js, NestJS, PostgreSQL, Socket.IO, Docker, and Git.
- Developed 40+ REST API endpoints supporting authentication, social feeds, messaging, notifications, moderation, analytics, and user management.
- Architected a PostgreSQL relational database containing 12 interconnected entities supporting scalable user, messaging, moderation, and analytics workflows.
- Implemented JWT authentication, RBAC authorization, secure file uploads, profile privacy controls, and backend security mechanisms.
- Developed real-time messaging, typing indicators, online presence tracking, and notification delivery using Socket.IO.
- Designed analytics dashboards, moderation queues, administrative tools, and platform management interfaces.
- Built 10+ application screens including Feed, Explore, Messaging, Analytics, Notifications, Profile Management, and Admin Dashboard.
- Independently designed, developed, deployed, and maintained the complete platform from architecture design through production deployment.

Nazma ERP — Commercial Business Management System

May 2026 – Present

- Leading development of a commercial ERP platform for Nazma Metal Industries focused on streamlining business operations and financial workflows.
- Designing and implementing 12 business modules covering sales, invoicing, collections, dealer management, ledger operations, reporting, and administration.
- Implementing accounting-sensitive workflows ensuring consistency across invoices, collections, balances, and financial records.
- Developing multilingual English and Bengali interfaces supporting operational users across multiple departments.
- Architecting RBAC systems supporting Super Admin, Manager, Accounts, and Sales Representative workflows.
- Applying Docker-based deployment strategies and modern software engineering practices throughout the software development lifecycle.

Education

American International University-Bangladesh (AIUB)

Expected Graduation: September 2026

Bachelor of Science (B.Sc.) in Computer Science Engineering

Major: Software Engineering — CGPA: 3.30 / 4.00

Research Publication

An AI-Driven Framework for Real-Time Fake News Detection: Developing a Machine Learning-Based Filter for News Platforms in the United States

Authors: Umma Twaha, Yeasin Arfin

Future Engineering Journal, 2025

Published peer-reviewed research exploring AI-driven fake news detection and machine learning-based content filtering systems for real-time news verification.

DOI: 10.54660/IJFEI.2025.2.4.158-169

Thesis

Stress Detection Using K-Nearest Neighbors (KNN)

- Conducting research on stress detection using machine learning techniques and behavioral data analysis.
- Utilizing Python, Pandas, NumPy, and machine learning algorithms for data preprocessing, feature engineering, model development, and evaluation.
- Applying classification techniques and statistical analysis to improve predictive model performance.